

Course Code: B23AM4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
Generative AI					
For AIML					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1	a).	Explain the key difference between Generative Models and Discriminative Models with an example.	1	2	2
	b).	Classify different types of Generative Models and mention one use case for each	1	2	2
	c).	Explain the role of attention mechanisms in Transformer architecture	2	2	2
	d).	Summarize the key differences between encoder-based models (e.g., BERT) and decoder-based models (e.g., GPT)	2	2	2
	e).	Explain the difference between GPT-4V and a text-only language model?	3	2	2
	f).	A user gives a text prompt to generate an image. Which model is more suitable: DALL·E 3 or VAE? Justify your answer.	3	3	2
	g).	A user wants to generate melody notes one after another. Determine which model is more suitable: RNN or CycleGAN? Justify your answer	4	3	2
	h).	Compare polyphonic music generation with simple melody generation?	4	2	2
	i).	Explain how Llama is useful in text generation	5	2	2
	j).	Describe any two uses of Google Copilot	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Apply the concept of Generative Adversarial Networks (GANs) in healthcare and Explain its working	1	3	5
	b).	Demonstrate how Variational Autoencoders (VAEs) can be used and Illustrate the generative process involved in it.	1	3	5
		OR			
3.	a).	Apply autoregressive models to a text generation task. Explain how the probabilistic modelling helps in generating coherent sequences.	1	3	5
	b).	Develop a use-case scenario where diffusion models (e.g., Vector	1	3	5

		Quantized Diffusion Models) can be applied. Explain the generative process and challenges involved			
		UNIT-2			
4.	a).	Apply the Transformer architecture to a text generation task. Explain how the encoder, decoder, and attention mechanisms work together to generate meaningful text.	2	3	5
	b).	Demonstrate how prompt engineering techniques can be used to improve responses from large language models. Provide examples and explain the role of Reinforcement Learning from Human Feedback (RLHF).	2	3	5
		OR			
5.	a).	Apply Retrieval-Augmented Generation (RAG) to design a system for Document question answering. Explain the workflow.	2	3	5
	b).	Demonstrate a use-case scenario where a multimodal LLM can be applied and discuss challenges such as hallucination.	2	3	5
		UNIT-3			
6.	a).	Explain how generator and discriminator compete for generating images during the adversarial training process in GANs.	3	2	5
	b).	A user wants to generate an image from the prompt: "A futuristic classroom in watercolor painting style." Determine whether Stable Diffusion or DALL·E 3 is more appropriate for this task, and justify your answer	3	3	5
		OR			
7.	a).	Describe the concept of Visual Transformers (ViT). How are images processed differently in ViT compared to CNN-based models?	3	2	5
	b).	Suppose a GAN model is repeatedly generating very similar images for different prompts. Using your knowledge of GANs, determine the issue and explain how you would address it.	3	3	5
		UNIT-4			
8.	a).	Explain the role of Recurrent Neural Networks (RNNs) in music generation. Why are they suitable for sequence modeling?	4	2	5
	b).	Analyze the advantages and limitations of CycleGAN in painting generation tasks	4	4	5
		OR			
9.	a).	Describe the architecture and purpose of MuseGAN in polyphonic music generation.	4	2	10
		UNIT-5			
10.	a).	Explain the purpose of Hugging Face AutoTrain in model training.	5	2	5

	b).	A user wants to generate text using an open-source model. Explain how Llama can be used for this purpose.	5	3	5
		OR			
11.	a).	Explain the role of LangChain in Generative AI applications	5	2	5
	b).	A student wants to run a Generative AI model on a laptop without internet. Determine whether GPT4All can be used or not, justify it.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



Course Code: B23HS4101					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
HUMAN RESOURCE & PROJECT MANAGEMENT					
(Common to CSE, AIML, IT, AIDS, CSG, CSIT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Write the role of HR manager.	1	1	2
	b).	Define recruitment.	1	1	2
	c).	List any two models of HR accounting.	2	1	2
	d).	Define career counseling.	2	1	2
	e).	Define Resource management.	3	1	2
	f).	State the 80-20 rule.	3	1	2
	g).	Define Brainstorming.	4	1	2
	h).	Write about Cash flow analysis.	4	1	2
	i).	List any two forms of project organization.	5	1	2
	j).	Define project control.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Interpret the nature and scope of Human Resource Management.	1	2	5
	b).	Discuss the emerging trends in HRM and their impact on organizations.	1	2	5
		OR			
3.	a).	Explain the ethical aspects of HRM with suitable examples.	1	2	5
	b).	Describe the steps involved in the selection procedure.	1	2	5
		UNIT-2			
4.	a).	Discuss the different methods of training used in organizations.	2	2	5
	b).	Explain the importance of performance appraisal in organizations.	2	2	5
		OR			
5.	a).	Summarize the concept of career development and its importance.	2	2	5
	b).	Explain group interaction and its importance in HRD.	2	2	5
		UNIT-3			
6.	a).	Explain the concept of Project Management and its importance.	3	2	5
	b).	Discuss the different types of projects with examples.	3	2	5

		OR			
7.	a).	Explain the stages of the project life cycle.	3	2	5
	b).	Interpret the process of project selection.	3	2	5
		UNIT-4			
8.	a).	Describe the criteria used for project selection.	4	2	5
	b).	A company invests ₹2,00,000 in a project. The expected cash inflows are: Year 1: ₹50,000 Year 2: ₹60,000 Year 3: ₹70,000 Year 4: ₹80,000 Calculate the Payback Period.	4	2	5
		OR			
9.		A project requires ₹2,00,000 investment. It generates ₹80,000 annually for 4 years. Discount rate = 10% Calculate the NPV and state whether the project is acceptable.	4	2	10
		UNIT-5			
10.	a).	Interpret the process of project planning in detail.	5	2	5
	b).	Explain the human aspects of project management.	5	2	5
		OR			
11.	a).	Discuss the prerequisites for successful project implementation.	5	2	5
	b).	Summarize the importance of project review.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23AM4102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
QUANTUM COMPUTING					
For AIML					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define quantum computing and briefly explain its significance.	1	2	2
	b).	Differentiate between bits and qubits with an example.	1	2	2
	c).	What is the role of linear algebra in quantum computing?	2	2	2
	d).	Explain the concept of superposition in quantum mechanics.	2	2	2
	e).	List two physical implementations of a qubit.	3	2	2
	f).	What is the function of a single qubit gate in a quantum circuit?	3	2	2
	g).	State the purpose of Deutsch's algorithm in quantum computing.	4	2	2
	h).	How does Grover's search algorithm provide an advantage over classical search methods?	4	2	2
	i).	Define quantum error correction and its importance in quantum computing.	5	2	2
	j).	What is quantum teleportation, and what are its key components?	5	2	2
5 x 10 = 50 Marks					
	UNIT-1				
2.	Analyze the importance of Mathematics, Physics, and Biology in the development and advancement of Quantum Computing. Provide relevant examples to support your analysis.		1	3	10
OR					
3.	a).	Compare and contrast Classical logical operations with Quantum logical operations, highlighting their fundamental differences.	1	2	5
	b).	Describe the basic principles and key concepts that form the foundation of Quantum Computing.	1	2	5
UNIT-2					
4.	a).	Explain the concept of measurements in bases other than the computational basis.	2	2	5
	b).	Discuss the significance and applications of such measurements in Quantum Computing.	2	2	5
OR					

5.	a).	Define Hilbert space and elaborate on its role in Quantum Mechanics and Quantum Computing.	2	2	5
	b).	Discuss the principles of entanglement and supersymmetry, and their implications in Quantum Computing.	2	2	5
UNIT-3					
6.	a).	Design a simple quantum circuit using single and multiple qubit gates.	3	3	5
	b).	Explain the function of each gate and the overall purpose of the circuit.	3	2	5
OR					
7.	a).	Illustrate the Bloch sphere and explain how it represents the state of a qubit.	3	2	5
	b).	Describe Bell states and their significance in quantum entanglement and quantum communication.	3	2	5
UNIT-4					
8.	a).	Examine Shor's factorization algorithm and analyze its impact on classical cryptography.	4	2	5
	b).	Discuss the potential consequences for security systems if large-scale quantum computers become a reality.	4	3	5
OR					
9.	a).	Explain the working of Deutsch's-Jozsa algorithm and its significance in demonstrating the power of quantum computation.	4	2	5
	b).	Describe Grover's search algorithm and discuss its applications and advantages over classical search algorithms.	4	3	5
UNIT-5					
10.		Analyze the principles of Quantum Cryptography and discuss its advantages over classical cryptographic methods. Provide examples of quantum cryptographic protocols.	5	3	10
OR					
11.	a).	Explain the concept of fault-tolerant computation in Quantum Computing and its importance.	5	2	5
	b).	Describe the process of quantum teleportation and discuss its significance in quantum communication and information transfer.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23AM4103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
CLOUD COMPUTING					
AI & ML					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	K L	M
1.	a).	What is cloud computing?	1	1	2
	b).	List any two advantages of cloud computing.	1	1	2
	c).	What is the role of Remote Procedure Call (RPC) in distributed systems?	2	1	2
	d).	Define parallel computing.	2	1	2
	e).	Mention one advantage and one disadvantage of virtualization	3	1	2
	f).	List any two characteristics of a virtualized environment.	3	1	2
	g).	Define cloud interoperability in cloud computing	4	1	2
	h).	Name two methods used to improve energy efficiency in cloud data centers.	4	1	2
	i).	Differentiate between serverless and traditional server-based computing	5	2	2
	j).	Name one public cloud and one open-source platform for serverless computing.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.		Explain the types of cloud service models with suitable example for each.	1	2	10
		OR			
3.		Explain the cloud computing reference model in detail, describing each abstraction layer and its function.	1	2	10
		UNIT-2			
4.		Explain the architecture types: SISD, SIMD, MISD, and MIMD. Provide examples of each.	2	2	10
		OR			
5.	a).	Discuss how Inter-Process Communication (IPC) and Remote Procedure Call (RPC) facilitate distributed computing.	2	2	5
	b).	Describe the role of virtualization, SOA, and web services as enabling	2	2	5

		technologies for the cloud. How do they support scalability and interoperability?			
		UNIT-3			
6.		Explain, with examples, the taxonomy of virtualization techniques (hardware, OS, server, network, storage, application virtualization)	3	2	10
		OR			
7.	a).	Describe in detail the working of Amazon Elastic Container Service and how it differs from VM-based services like EC2.	3	2	5
	b).	Briefly describe the taxonomy of virtualization techniques, mentioning Type 1 and Type 2 hypervisors and their key differences.	3	2	5
		UNIT-4			
8.		Explain how cloud scalability and fault tolerance improve service availability for enterprise applications.	4	2	10
		OR			
9.	a).	Explain the concept of federated clouds. What challenges do they address and introduce?	4	2	5
	b).	Describe the cloud security architecture. Discuss how security varies across different deployment models (public, private, hybrid).	4	2	5
		UNIT-5			
10.	a).	Explain how event-driven execution works in serverless platforms.	5	2	5
	b).	Describe the architecture of Function-as-a-Service (FaaS) with examples such as AWS Lambda and OpenFaaS.	5	2	5
		OR			
11.		Discuss the role of edge and fog computing in extending the benefits of serverless architectures	5	2	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

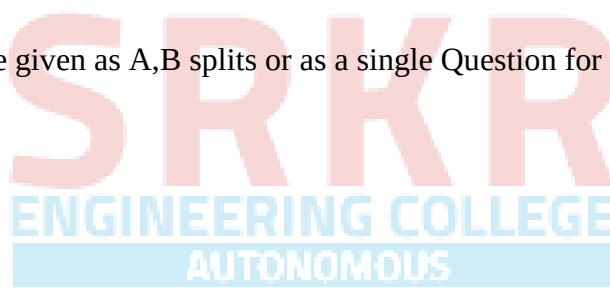
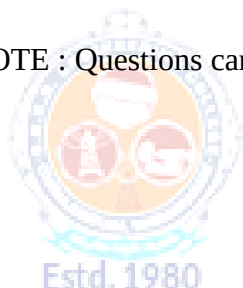
NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CS4103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
BLOCKCHAIN TECHNOLOGY					
(Common to CSE & AIML)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the origin of blockchain.	1	2	2
	b).	Explain different types of blockchain.	1	2	2
	c).	Explain public blockchain.	2	2	2
	d).	Explain different types of Oracles	2	2	2
	e).	Explain Byzantine Fault.	3	2	2
	f).	Discuss the advantages of Ripple.	3	2	2
	g).	Explain Security Aspects in Bitcoin	4	2	2
	h).	Explain Limitations of Blockchain	4	2	2
	i).	Discuss How to Use Python Online Editor	5	2	2
	j).	Explain the Chain codes.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the Components of Blockchain.	1	2	5
	b).	Identify the importance of Consensus Protocol in the Blockchain.	1	2	5
		OR			
3.	a).	Explain Decentralization and Distribution in the Blockchain.	1	2	5
	b).	Explain different types of Cryptocurrencies.	1	2	5
		UNIT-2			
4.	a).	Explain Public Blockchain system.	2	2	5
	b).	Discuss different Characteristics of a Smart Contract	2	2	5
		OR			
5.	a).	Explain the Ethereum Blockchain system.	2	2	5
	b).	Explain Smart Contracts in Ethereum.	2	2	5
		UNIT-3	3	2	5
6.	a).	Explain Key Characteristics of Private Blockchain.	3	2	5
	b).	Discuss different Blockchain Fundraising Methods.			

		OR			
7.	a).	Explain Characteristics of Consortium Blockchain	3	2	5
	b).	Discuss Pros and Cons of Initial Coin Offering	3	2	5
		UNIT-4			
8.	a).	Explain the Security and Privacy Challenges of Blockchain.	4	2	5
	b).	Illustrate how Blockchain can be used in Banking and Finance.	4	3	5
		OR			
9.	a).	Discuss the process of Safeguarding Blockchain Smart Contract.	4	2	5
	b).	Illustrate how Blockchain can be used in Internet of Things	4	3	5
		UNIT-5			
10.	a).	Illustrate how Python Packages can be used for Blockchain.	5	3	5
	b).	Illustrate Blockchain Application development Using Fabric Java SDK.	5	3	5
		OR			
11.	a).	Explain different Components of Hyperledger Fabric Network	5	2	5
	b).	Illustrate how Blockchain can be used in Health Care.	5	3	5

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



Course Code: B23AM4104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
SOFTWARE PROJECT MANAGEMENT					
(For AIML)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	KL	M
1.	a)	What is meant by software economics?	1	2	2
	b)	What is meant by transitioning to an iterative process?	1	2	2
	c)	What is the primary objective of the Elaboration phase?	2	2	2
	d)	What are engineering artifacts? Provide two examples.	2	2	2
	e)	How do software models support communication between stakeholders?	3	2	2
	f)	What is a Work Breakdown Structure (WBS) in project planning?	3	2	2
	g)	What is the primary role of a Line-of-Business (LOB) organization in a project?	4	2	2
	h)	Differentiate between quality indicators and management indicators.	4	2	2
	i)	What is the core principle of Agile methodology?	5	2	2
	j)	What is meant by "orchestration" in a DevOps context?	5	2	2
5 x 10 =50Marks					
		UNIT-1			
2.	a)	Discuss the limitations of the Waterfall model in managing complex software projects. How does it affect project performance?	1	3	5
	b)	Compare traditional and pragmatic software cost estimation methods with examples.	1	3	5
		OR			
3.	a)	Apply strategies for improving team effectiveness to achieve software project success with practical examples. .	1	3	5
	b)	What are the challenges involved in transitioning from a traditional to an iterative development process? How can they be overcome?	1	3	5
		UNIT-2			
4.	a)	Analyse the role of the Construction phase in software development. How does it contribute to product stability and readiness?	2	3	5

	b)	Apply suitable strategies to address the key challenges faced during the Transition phase for ensuring smooth software deployment.	2	3	5
		OR			
5.	a)	Compare and contrast Engineering artifacts and Programmatic artifacts in terms of their roles and examples.	2	3	5
	b)	Discuss how effective artifact management can improve project transparency and collaboration among stakeholders.	2	3	5
		UNIT-3			
6.	a)	Compare the management and technical perspectives in model-based software architectures. Give examples of their respective concerns.	3	3	5
	b)	Describe the role of iteration workflows in modern software development. How do they differ from traditional workflows?	3	3	5
		OR			
7.	a)	Discuss the importance of periodic status assessments in an iterative software process. How do they help mitigate risks?	3	3	5
	b)	Describe the steps involved in iteration planning. How does it support timely delivery and continuous feedback?	3	3	5
		UNIT-4			
8.	a)	Discuss the roles and responsibilities of Line-of-Business (LOB) organizations versus dedicated Project Organizations. How do these entities collaborate and sometimes conflict within complex projects?	4	3	5
	b)	Discuss the impact of process standardization and automation on quality assurance and productivity in software projects. How does automation influence the role of project managers and team members?	4	3	5
		OR			
9.	a)	Discuss the significance of the seven core metrics in project control. How can these metrics be effectively used to predict project risks, improve management decisions, and ensure quality delivery?	4	3	5
	b)	Discuss the concept of life cycle expectations in software project metrics. How do these expectations guide the selection and interpretation of pragmatic software metrics throughout the project life cycle?	4	3	5
		UNIT-5			
10.	a)	Describe the role of Scrum patterns in overcoming challenges during Agile adoption. How can these patterns help teams iterate towards greater agility?	5	3	5
	b)	Describe the architecture of a typical DevOps environment. How do deployments and orchestration fit into this architecture?	5	3	5

		OR			
11.	a)	Describe the importance of orchestration in managing instances of applications in a cloud environment. Provide an example of how orchestration tools improve deployment efficiency.	5	3	5
	b)	Discuss how Agile capabilities support the adoption of DevOps in an organization. What changes in team processes and mindset are necessary?	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23CS4107					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
AGILE METHODOLOGIES					
(Common to CSE,AIIML, AIDS, CSBS, IT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a)	What is meant by Agile manifesto	1	1	2
	b)	Define task board	1	1	2
	c)	Define self organizing team	2	1	2
	d)	Suggest appropriate agile communication principles for a software development team that is struggling with communication gaps and delayed deliveries in Ebook Reader project.	2	3	2
	e)	Mention the three main roles on a Scrum project.	3	1	2
	f)	How is Scrum both incremental and iterative?	3	2	2
	g)	Define pair programming	4	1	2
	h)	What is feedback loops and why do XP teams create them	4	2	2
	i)	Define Shuhari	5	1	2
	j)	What are the types of waste that create constraints in the workflow?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a)	Explain the core values of Agile manifesto and describe how they differ from traditional software development approaches	1	2	5
	b)	Analyze how a “fractured perspective” leads to project failure. How does Agile thinking help resolve this issue?	1	3	5
		OR			
3.	a)	Explain the idea of “Principles over Practices” in Agile. Why is focusing only on practices without understanding underlying principles often ineffective	1	2	5
	b)	Illustrate with an example how “Responding to Change over Following a Plan” benefits a software project.	1	3	5
		UNIT-2			
4.		Explain Agile principles and their importance in software development.	2	2	10
		OR			
5.	a)	Explain how effective communication contributes to better project delivery in software development.	2	2	5

	b)	Explain how continuous improvement contributes to better project and team performance in Agile	2	2	5
		UNIT-3			
6.	a)	Explain the Scrum framework. Discuss the roles of Product Owner, Scrum master, and development team	3	2	5
	b)	Design a complete sprint planning process for a mobile application development project. Apply user stories, story points. Velocity estimation and task board management to create a comprehensive sprint plan.	3	3	5
		OR			
7.	a)	Explain the significance of Scrum values. Explain why Scrum practices alone are insufficient without adherence to these values.	3	2	5
	b)	A sprint fails because the Product Owner frequently changes priorities during sprint. Use Scrum planning rules to analyze the product owner support for sprint success without disrupting execution.	3	3	5
		UNIT-4			
8.		Explain the XP values and their role in shaping an effective development mindset	4	2	10
		OR			
9.	a)	Illustrate how continuous integration helps in maintaining software quality in XP	4	2	5
	b)	Explain how XP practices help teams embrace changing requirements effectively.	4	2	5
		UNIT-5			
10.	a)	Explain Lean thinking and identify different types of waste in software development	5	2	5
	b)	Analyse how limiting Work in Progress (WIP) improves flow in a Kanban system	5	3	5
		OR			
11.	a)	Explain Kanban principles to manage workflow in a software maintenance project	5	2	5
	b)	Evaluate the role of Agile Coach in helping organizations adopt agile practices successfully	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23AM4106					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
HIGH PERFORMACE COMPUTING					
Common to AIML & AIDS					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is the Scope of Parallel Computing?	1	1	2
	b).	How does N-wide superscalar architecture improves HPC performance?	1	2	2
	c).	Mention any two Decomposition Techniques in Parallel Computing?	2	1	2
	d).	What is GPU in computing?	2	1	2
	e).	What is the need of All-Reduce operation in HPC?	3	1	2
	f).	What is the primary goal of OpenMP programming in HPC	3	1	2
	g).	What is the the effect of granularity on performance analysis in hpc	4	1	2
	h).	What does it mean when we say a matrix is dense?	4	1	2
	i).	Mention any two Issues in Sorting on Parallel Computers?	5	1	2
	j).	List any two common examples of CUDA C kernels used in parallel computing?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Define Implicit Parallelism?	1	1	3
	b).	Explain in brief about Trends in Microprocessor and Architectures?	1	2	7
		OR			
3.	a).	Write a note on Communication costs in parallel machines?	1	1	5
	b).	Describe Multi-core architecture in detail?	1	2	5
		UNIT-2			
4.	Explain exploratory Decomposition and Hybrid Decomposition techniques with an example?		2	3	10
		OR			
5.	a).	What are the characteristics of tasks?	2	2	4
	b).	Describe mapping technique for load balancing?	2	2	6

		UNIT-3			
6.	a).	Explain prefix-sum operation for an eight-node hypercube?	3	2	6
	b).	Describe scatter and gather in detail?	3	2	4
		OR			
7.	a).	Explain All-to-One Broadcast and Reduction on a ring?	3	2	5
	b).	Explain with example and algorithm all-to-all broadcast on 3×3 mesh?	3	2	5
		UNIT-4			
8.	a).	Explain performance matrices of parallel system.	4	2	5
	b).	Write a note on minimum and cost optimal execution time.	4	2	5
		OR			
9.	Explain the Dense matrix algorithms. a). Matrix vector multiplication. b). Matrix - Matrix multiplication.		4	3	5
			4	3	5
		UNIT-5			
10.	a).	Explain parallel Depth- First search in details?	5	3	5
	b).	Illustrate with an example bubble sort and its variants?	5	2	5
11.	a).	Describe the processing flow of CUDA C program with a diagram?	5	2	7
	b).	What is the main purpose of CUDA in programming?	5	1	3

CO-COURSE OUTCOME KL-KNOWLEDGE LEVEL M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23AM4107					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
Explainable AI					
(For AIML)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a)	What is meant by interpretability in AI models?	1	2	2
	b)	What is a counterfactual explanation?	1	2	2
	c)	What are interpretable models in machine learning?	2	2	2
	d)	What is meant by symbolic AI?	2	2	2
	e)	Define Integrated Gradients.	3	2	2
	f)	Define Partial Dependence Plot (PDP)	3	2	2
	g)	What are regulatory considerations in AI?	4	2	2
	h)	Define quantitative metrics for interpretability.	4	2	2
	i)	What is the role of XAI in healthcare applications?	5	2	2
	j)	What is the need for explainability in self-driving cars?	5	2	2
5 x 10 =50Marks					
		UNIT-1			
2.	a)	Explain the motivations for Explainable AI (XAI) with suitable examples.	1	2	5
	b)	Explain how feature importance techniques are used for post-hoc explanations.	1	2	5
		OR			
3.	a)	Explain how rule-based systems can be used for explainable AI with an example.	1	2	5
	b)	Distinguish model-specific and model-agnostic interpretability methods.	1	2	5
		UNIT-2			
4.	a)	Apply a decision tree to solve a classification problem and explain its interpretability.	2	3	5
	b)	Illustrate the working of sparse neural networks and how sparsity improves interpretability.	2	3	5
		OR			

5.	a)	Apply attention mechanisms to interpret predictions in deep learning models.	2	3	5
	b)	Describe the process of Layer-wise Relevance Propagation with an example.	2	3	5
		UNIT-3			
6.	a)	Illustrate the working of LIME with an example.	3	3	5
	b)	Illustrate the concept of Conformal Prediction with an example.	3	3	5
		OR			
7.	a)	Illustrate the working of SHAP with an example.	3	3	5
	b)	Apply saliency maps to interpret predictions of a deep learning model.	3	3	5
		UNIT-4			
8.	a)	Explain different quantitative metrics used to evaluate interpretability with examples.	4	2	5
	b)	Explain the ethical implications of Explainable AI with real-world scenarios.	4	2	5
		OR			
9.	a)	Explain human-centric evaluation methods to assess an XAI system.	4	2	5
	b)	Explain how accountability can be ensured in AI-based decision systems.	4	2	5
		UNIT-5			
10.	a)	Demonstrate how XAI can be used for fraud detection systems with an example.	5	3	5
	b)	Demonstrate how explainability techniques are used to interpret predictions in drone-based systems.	5	3	5
		OR			
11.	a)	Discuss the importance of XAI in time series forecasting with an example.	5	3	5
	b)	Demonstrate how explainability is achieved in Computer Vision models.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23AM4108					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
Agentic AI					
(For AIML)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is the difference between Agentic AI and Generative AI?	1	2	2
	b).	Define rationality in intelligent agents	1	2	2
	c).	What is meant by planning under uncertainty?	2	2	2
	d).	What is adaptation in intelligent agents?	2	2	2
	e).	Define prompt engineering	3	1	2
	f).	What is the role of LLMs in agentic systems	3	2	2
	g).	What is an embedding database	4	1	2
	h).	What is meant by deployment of AI agents?	4	2	2
	i).	Define interpretability in AI	5	1	2
	j).	What is the role of AI in cybersecurity?	5	2	2
5 x 10 =50Marks					
		UNIT-1			
2.	a).	Describe the characteristics of Agentic AI such as autonomy, proactivity, and goal-directed behavior	1	2	5
	b).	Describe different agent architectures with suitable examples.	1	3	5
		OR			
3.	a).	Explain the components of an intelligent agent with a neat diagram	1	2	5
	b).	Discuss real-world applications of Agentic AI	1	3	5
		UNIT-2			
4.	a).	Explain utility theory and its role in decision-making.	2	2	5
	b).	Explain reward shaping with an example.	2	3	5
		OR			
5.	a).	Describe the fundamentals of reinforcement learning, including policy, reward, and value function..	2	2	5
	b).	Explain how agents adapt to dynamic environments using learning techniques	2	3	5

		UNIT-3			
6.	a).	Describe the role of Large Language Models (LLMs) in agentic systems.	3	2	5
	b).	Explain different prompt engineering techniques such as zero-shot, chain-of-thought, and ReAct.	3	3	5
		OR			
7.	a).	Discuss the importance of context management and embeddings in LLM-based systems.	3	2	5
	b).	Describe the working of RAG with an example application	3	3	5
		UNIT-4			
8.	a).	Discuss how agents communicate with each other in a multi-agent environment.	4	2	5
	b).	Compare different agent frameworks based on their features and use cases?	4	3	5
		OR			
9.	a).	Explain key production concerns such as reliability, latency, and scalability in agentic AI.	4	2	5
	b).	Explain how multi-agent workflows are designed using modern frameworks?	4	3	5
		UNIT-5			
10.	a).	Discuss different risk mitigation strategies in agentic AI systems.?	5	2	5
	b).	Discuss the application of agentic AI in healthcare systems	5	3	5
		OR			
11.	a).	Describe the importance of privacy, transparency, and accountability in AI systems	5	2	5
	b).	Analyze a real-world case study of responsible AI implementation	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23AM4109					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
Edge AI					
(For AIML)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Describe any two key techniques used in Edge Computing.	1	2	2
	b).	Explain the term Intelligent Edge.	1	2	2
	c).	Explain the concept of virtualization in computing systems.	2	2	2
	d).	Explain the concept of Continuous Integration	2	2	2
	e).	What is an edge device in AI-based systems	3	2	2
	f).	What is NVIDIA TensorRT	3	2	2
	g).	What is meant by Adaptive Edge Caching?	4	2	2
	h).	What is meant by Edge Task Offloading?	4	2	2
	i).	What is meant by edge inference in Mobile Edge AI	5	2	2
	j).	What is model partition-based edge training?	5	2	2
5 x 10 =50Marks					
UNIT-1					
2.	a).	Explain the fundamentals of Edge Computing and its importance in modern distributed systems.	1	2	5
	b).	Describe the key techniques used in Edge Computing with suitable examples	1	2	5
OR					
3.	a).	Explain the industrial applications of Edge Computing in areas such as manufacturing, healthcare, and smart cities.	1	2	5
	b).	Explain the opportunities and future scope of Edge Computing.	1	2	5
UNIT-2					
4.	a).	Describe the high-level hardware hierarchy of the Edge Computing paradigm	2	2	5
	b).	Apply the process of configuring a CI/CD pipeline using GitLab.	2	3	5
OR					
5.	a).	Explain the benefits and challenges of DevOps implementation.	2	2	5
	b).	Demonstrate the use of network virtualization in cloud or edge environments	2	3	5

		UNIT-3			
6.	a).	Apply AI model optimization techniques to deploy a deep learning model on an edge device.	3	3	5
	b).	Demonstrate the process of converting a trained AI model into ONNX format	3	3	5
		OR			
7.	a).	Illustrate how AI model segmentation can improve performance in Edge AI systems.	3	3	5
	b).	Apply the concept of sharing AI computation among multiple edge nodes	3	3	5
		UNIT-4			
8.	a).	Illustrate how AI can be applied for adaptive edge caching in real-time applications.	4	3	5
	b).	Demonstrate the use of Deep Reinforcement Learning (DRL) for optimizing edge resource allocation	4	3	5
		OR			
9.	a).	Demonstrate how AI can improve security management in Edge Computing systems.	4	3	5
	b).	Illustrate how AI-based predictive maintenance can improve reliability of edge devices.	4	3	5
		UNIT-5			
10.	a).	Demonstrate how computation offloading improves performance in Mobile Edge AI systems.	5	3	5
	b).	Apply coded computing techniques to improve reliability in distributed edge AI systems.	5	3	5
		OR			
11.	a).	Design a Mobile Edge AI system architecture for real-time applications.	5	3	5
	b).	Apply agile capabilities in the development of Mobile Edge AI solutions.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARK

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks